Cap Torque Specification and Inspection Method

Purpose and Scope

This specification defines the required application torque for screw closures on Riverbend Packaging bottles and the method used to verify it. Correct torque protects against leakers, tamper-band damage, and consumer complaints from caps that are too tight to open. It applies to all products run on the CX-4 capper.

Torque Requirements

Application torque is measured as removal torque 24 hours after capping unless otherwise stated. The specification limits below apply to the measured value on a calibrated digital torque tester.

Cap size	Bottle	Specification	Action limits
38 mm	500 mL	**18 to 22 in-lb**	Below 17 or above 23 stops line
28 mm	250 mL	12 to 16 in-lb	Below 11 or above 17 stops line
33 mm	1 L	16 to 20 in-lb	Below 15 or above 21 stops line

NOTE: The specification applies to the finished package at the end of the line. Capper settings must be adjusted so that finished-package torque falls inside this range.

Sampling Plan

Production state	Sample size	Frequency
Startup or after changeover	10 bottles	Before release
Normal production	5 bottles	Every 30 minutes
After a jam or restart	5 bottles	Immediately
Suspect lot	20 bottles	As directed by Quality

Sampled bottles are taken from all four capper heads, with at least one from each head, and labeled with head number.

Inspection Method

1. Verify the torque tester has a current calibration sticker. Do not use an expired tester.
2. Zero the tester with no bottle installed.
3. Place the bottle in the fixture and secure the closure in the tester chuck.
4. Rotate the closure counterclockwise at a steady rate until the cap breaks free.
5. Record the peak value in the SPC log.
6. Repeat for the remaining samples.

7 Compute the average and range of the sample.

8 Compare the results to the specification and action limits.

Disposition of Results

Result	Action
All values within specification	Continue production
Any value within spec but average trending toward a limit	Notify maintenance for adjustment
One or more values outside spec but inside action limits	Resample 10 bottles; adjust capper
Any value outside action limits	Stop the line, notify Quality, hold product since last good check

Product held for torque is handled under the Nonconformance and Hold Procedure. The Quality Engineer determines whether the product is reworked, sorted, or scrapped.

Visual Inspection

In addition to torque, inspect for cocked caps, cracked tamper bands, scuffed threads, and missing liners. Reject any bottle with a cap not seated flush on the neck finish.

Equipment and Calibration

Torque testers are calibrated annually by an external lab and verified monthly with a reference weight. Record the verification in the equipment log. A tester that fails verification is tagged out of service and all results since the last good verification are reviewed by the Quality Engineer.

Relationship to Capper Settings

The capper clutch is a means to achieve the finished-package torque, not the requirement itself. Differences between clutch setting and finished torque occur because of liner relaxation and thread friction. Maintenance adjusts settings based on quality data, and Quality confirms the effect with the sampling plan above.

Records

SPC charts and torque logs are retained for two years and reviewed weekly by the Quality Engineer. Out-of-specification events are reported at the weekly quality meeting.